Howard University
College of Engineering & Architecture
Department of Electrical Engineering & Computer Science

EECE 306-01: Fundamentals of Electromagnetics Laboratory

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Lab 8: Dielectrics, Boundary Conditions, and Capacitance

Midterm milestone. This lab carries a formal report in the department nine section format, which replaces this week's notebook.

Objectives

By the end of this session you will have

- extended the solver to handle material interfaces correctly,
- shown numerically why the naive averaging of permittivity is wrong,
- computed capacitance and resistance from stored energy,
- obtained a number the textbook cannot give you in closed form.

1. Where the first half of the course has been going

Seven weeks of building arrive here. Today the toolbox reaches 43 of its 53 functions and computes, for the first time, a quantity that has no elementary closed form.

That last point deserves emphasis because it is the whole justification for the effort. Anyone can look up the capacitance of a coaxial line. Nobody can look up the capacitance of the particular microstrip geometry sitting on the board in front of them, and the ability to compute it is what separates an engineer from a person who owns a formula sheet.

2. Part I. Dielectrics in the solver

With a spatially varying permittivity the governing equation is

$$\nabla \cdot (\varepsilon \nabla V) = -\rho_v$$

and the permittivity does not come outside the divergence.

Discretizing in flux conserving form, the coefficient linking node (i, j) to its neighbour across a face is the permittivity *at that face*, not at either node. The question is what value to use when the two adjacent cells have different permittivity.

2.1 Harmonic, not arithmetic

The correct face value is the **harmonic mean**,

$$\varepsilon_{\text{face}} = \frac{2\varepsilon_1 \varepsilon_2}{\varepsilon_1 + \varepsilon_2}$$

The reason is a boundary condition you already know. Across a charge free interface the normal component of $\vec{\boldsymbol{D}}$ is continuous while the normal derivative of V is *discontinuous*, since

$$\varepsilon_1 \frac{\partial V}{\partial n}\bigg|_1 = \varepsilon_2 \frac{\partial V}{\partial n}\bigg|_2$$

Two resistive elements in series carry the same flux and their potential drops add, which is precisely the structure that produces a harmonic mean. An arithmetic mean would correspond to the two elements sharing a potential drop, which is not what happens.

2.2 The required study

Implement both choices behind a flag. Solve a two layer parallel plate capacitor, for which the exact answer is two capacitors in series and is elementary. Run a grid refinement study with each choice and report the observed convergence order for both.

The harmonic mean retains second order. The arithmetic mean degrades, because the interface error does not shrink at the same rate as the interior error. Report both numbers. This is a case where the wrong choice still produces a plausible answer on any single grid, and only the convergence study exposes it.

A plausible answer is the dangerous kind

An implementation that is simply broken announces itself. An implementation that is subtly wrong returns a number in the right range, with the right units, that moves in the right direction when you change the geometry.
The only reliable defence is to check the *rate* at which the answer improves, not just the answer. That is why every lab in this course asks for a convergence study rather than a single number, and today is the clearest illustration of why.

3. Part II. Energy, capacitance, and resistance

3.1 `em.derive.energyE(G, V, epsr)`

$$W = \frac{1}{2} \int \varepsilon \, |\nabla V|^2 \, \mathrm{d}A$$

per unit length for a two dimensional cross section. Use `em.op.grad` on the interpolated solution, or the discrete gradient on the grid, and state which you chose.

3.2 `em.derive.capacitance(G, epsr, mask, Vfix, opts)`

Solve with one conductor at V_0 and the other at zero, compute the stored energy, and

$$C = \frac{2W}{V_0^2}$$

The alternative route counts the surface charge on a conductor by integrating $\vec{\boldsymbol{D}}$ over a surface enclosing it. That route is legitimate and it is worse. It requires a surface that hugs an irregular conductor boundary, and on a coarse grid the staircased surface introduces an error that the energy route does not have. The API mandates the energy method for this reason.

3.3 `em.derive.resistance(G, sigma, mask, Vfix, opts)`

The steady current problem obeys

$$\nabla \cdot (\sigma \nabla V) = 0$$

which is the same equation with σ in place of ε. Your solver needs no change at all, only a different coefficient array. Compute the total current by integrating the current density over a surface and take $R = V_0/I$.

Point out in your report the correspondence between the two problems, which is why the per unit length product RC of a uniform two conductor geometry depends only on the material properties and not on the shape.

4. Part III. The validation ladder

Three rungs, in order. Do not skip to the top.

4.1 Rung one, parallel plate

$$C = \frac{\varepsilon A}{d} \quad \text{per unit area, neglecting fringing}$$

This catches gross errors, a missing factor of two, a wrong permittivity, a sign. Use a plate separation small compared with the plate width so that fringing is negligible, and state how small you needed it to be before the agreement was acceptable.

4.2 Rung two, coaxial line

$$C' = \frac{2\pi\varepsilon}{\ln(b/a)} \quad \text{per unit length}$$

This is a real test, because the geometry is curved and your grid is Cartesian, so the circular conductor boundary is staircased. Run a grid refinement study and report the observed convergence order.

Expect the order to be lower than 2. A staircased curved boundary is a first order representation of the geometry no matter how good the interior stencil is. Report what you measure and explain it. This is not a bug, it is a known limitation of Cartesian grids and it is why production solvers use body fitted or unstructured meshes.

4.3 Rung three, microstrip

A conducting strip of width w above a ground plane, separated by a substrate of thickness h and relative permittivity ε_r, with air above.

There is no elementary closed form for this. Approximate formulas exist in the microwave literature and they are fits, not derivations. Your solver computes it directly.

Sweep w/h across a decade at fixed ε_r, produce the per unit length capacitance curve, and from it the characteristic impedance using the standard relation between Z_0, C' and the capacitance of the same geometry with the substrate replaced by air.

Why the ladder is in that order

A number produced by a solver that has never been validated is not a result, it is a rumor. Rungs one and two are what earn you the right to believe rung three.
This ordering is not academic caution. It is the actual working procedure in any group that

computes for a living, and reviewers of your future papers will ask for exactly this ladder. Present it in your report in this order, explicitly labelled.

Required tests

Write `tests/test_lab08.m` containing at least these checks.

1. Parallel plate capacitance within one percent of $\varepsilon A/d$ at your stated aspect ratio.
2. Two layer parallel plate matches the series combination, with the harmonic mean, to within one percent.
3. Harmonic mean retains observed order near 2 while arithmetic mean does not. Report both.
4. Coaxial per unit length capacitance within two percent of $2\pi\varepsilon/\ln(b/a)$ on your finest grid.
5. Capacitance is independent of the applied V_0, checked at $V_0 = 1$ and $V_0 = 100$. This catches a missing square in the energy formula.
6. Filling a capacitor with ε_r multiplies its capacitance by ε_r.
7. Resistance of a uniform bar matches $L/(\sigma A)$.
8. The tangential $\vec{\boldsymbol{E}}$ is continuous across a dielectric interface, sampled on both sides.
9. The full suite from Lab 1 through Lab 8 passes in one run.

Deliverables

This is the milestone, so the deliverables differ from a normal week.

1. **Formal report**, in the department nine section format, covering the work of Labs 1 through 8 as a whole rather than this week alone. It must present the validation ladder explicitly and include the microstrip design curve. The report replaces the weekly notebook this week.
2. **Toolbox snapshot**, `Team`*NN*`_Lab08.zip`, now at 43 of 53 functions.
3. **Full regression suite** passing from Lab 1 through Lab 8.

Looking ahead

Next week the toolbox turns to magnetostatics, and you will find that most of the work is already done. Read Sections 3.1 and 3.2 of Wentworth before you come. Come with `em.field.E` and `em.quad` in good order, because next week rewards a clean implementation and punishes a tangled one.